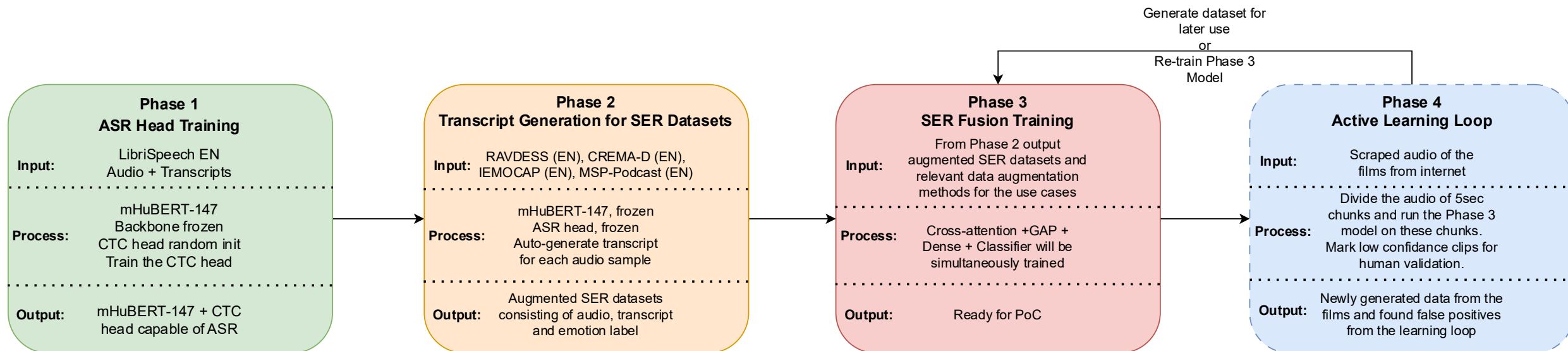


CLEAR

Proof of Concept – Overview

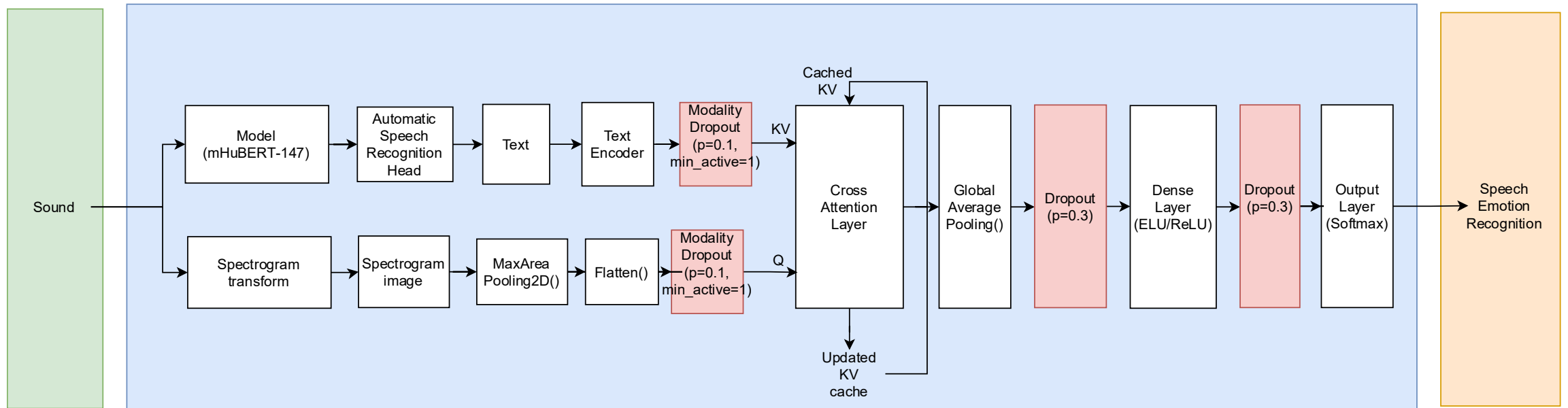
Date: Friday - 09/07/2026



Inputs

Proposed Multimodal Approach

Outputs



Phase 1 - mHuBERT-147 CTC ASR results

Configuration	val-CER	val-WER
Baseline (single layer 9)	8.2%	~31%
Weighted-sum [6-12]	5.17%	~19.1%
Weighted-sum [8,9,10]	4.4%	~17%
Weighted-sum [9,10]	4.4%	~16.7%
LoRA [9,10]	3.37%	~12.2%

Phase 2 - Emotion Datasets

Dataset	Clips	Hours
FOIA 911 (caller(?))	13,462	72.7h
ASVP-ESD	13,802	17.8h
CREMA-D	7,442	5.3h
RAVDESS	1,440	1.7h
TESS	2,800	1.6h
JL-Corpus	2,396	1.4h
SAVEE	480	0.5h
Kaggle Emergency	338	0.3h
Total	42,160	101.3h

Label	Hours	Hours %
panic	31.0h	30.6%
neutral	26.3h	25.9%
urgency	19.9h	19.6%
distress	17.3h	17.1%
confusion	3.5h	3.5%
fear	3.4h	3.4%
Total	101.3h	100%

Augmentation/Background Noise

Dataset	Hours	Use in pipeline
MUSAN	~6h (noise subset)	Babble, white/pink/brown noise, fan, HVAC
DEMAND	~6h	Real environments, car, bus, cafeteria, street traffic
RIR (OpenSLR 28)	~14h	Room reverb (used sparingly, phone audio is dry)
FSDnoisy18k	~42.5h	Ambient sound events (curated to ~15h safe subset)
UrbanSound8K	~9h	Supplemental urban ambient (siren/gunshot included)

Method	Domain	Purpose
Telephone codec simulation	Waveform	Studio → phone channel (8kHz, GSM / AMR / G.711) — closes the train↔test domain gap
Background noise injection	Waveform	Real 911 acoustics at 5-20dB SNR, emotion-uncorrelated noise pool only
SpecAugment	Log-mel	Time/freq masking for partial-signal robustness
Simplex distortion	Waveform	Speed / pitch / volume — speaker & recording variation
RIR reverb	Waveform	Light room reverb (phone calls are near-dry)